where $\rho(g) \in \mathrm{End}V$ is induced by $g$. The action of $L(\mathbf{x})$ on an element $h \in G$ is

$$L(\mathbf{x})h = \sum_{g\in G} x_g \rho(g)h = \sum_{g\in G} x_g gh = \sum_{z\in G} x_{zh^{-1}} z.$$

So the matrix of the linear operator $L(\mathbf{x})$ in the basis $g_1, g_2, \ldots, g_n$ is $X_G$ with permuted columns and hence has the same determinant up to sign.

Further, by Maschke's theorem, we have

$$\det_V L(\mathbf{x}) = \prod_{i=1}^{r} (\det_{V_i} L(\mathbf{x}))^{\dim V_i},$$

where $V_i$ are the irreducible representations of $G$. We set $P_i = \det_{V_i} L(\mathbf{x})$. Let $\{e_{im}\}$ be bases of $V_i$ and let $E_{i,jk} \in \mathrm{End}\, V_i$ be the matrix units in these bases. Then $\{E_{i,jk}\}$ is a basis of $\mathbb{C}[G]$ and

$$L(\mathbf{x})|_{V_i} = \sum_{j,k} y_{i,jk} E_{i,jk},$$

where $y_{i,jk}$ are new coordinates on $\mathbb{C}[G]$ related to $x_g$ by a linear transformation. Then

$$P_i(\mathbf{x}) = \det|_{V_i} L(\mathbf{x}) = \det(y_{i,jk}).$$

Hence, $P_i$ are irreducible (by Lemma 4.10.3) and not proportional to each other (as they depend on different collections of variables $y_{i,jk}$). The theorem is proved. □

## 4.11. Historical interlude: Georg Frobenius's "Principle of Horse Trade"

Ferdinand Georg Frobenius (1849–1917) studied at the famous Berlin University under both Karl Weierstrass and Leopold Kronecker, two great mathematicians who later became bitter opponents. Weierstrass considered Frobenius one of his brightest doctoral students and greased the wheels of his career by securing him a full professorship at the Zürich Polytechnikum. In Zürich Frobenius quickly got married, but the joys of happy matrimony did not prevent him from continuing productive research. Frobenius earned a high reputation for his